

Design of Reconfigurable MIMO Antenna with Enhanced Performance for 5G Application

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Abstract. This paper presents the design and performance analysis of a compact, reconfigurable 8×8 superwideband (SWB) multiple-input–multiple-output (MIMO) antenna designed for high-capacity wireless communication applications. Featuring a footprint of $55\text{mm} \times 55\text{ mm} \times 1.6\text{ mm}$ (corresponding to $0.367 \lambda_0 \times 0.367 \lambda_0 \times$ at the lowest operating frequency), the antenna utilizes microstrip line-fed half-circular radiating elements arranged with rotational symmetry on an FR-4 substrate. To enhance impedance bandwidth and inter-element isolation, a modified connected partial ground plane with a rectangular slot is integrated into the design. To introduce switchable frequency characteristics, a T-shaped stub is connected to the partial ground plane via a cadmium sulphide (CdS)-based photoconductive semiconductor switch. Under optical stimulation from a light-emitting diode (LED), the antenna's operating characteristics undergo a distinct transformation: in the switch-off (unilluminated) state, the antenna achieves a reconfigurable band of 8–16 GHz, whereas in the switch-on (illuminated) state, it delivers a superwideband performance spanning 2–20 GHz (10:1 ratio), effectively covering multiple bands including the sub-6 GHz region. Due to the symmetrical arrangement and orthogonal orientation of the elements, the antenna achieves stable radiation patterns, high port isolation, and excellent pattern diversity. The diversity performance is rigorously evaluated using key MIMO parameters, demonstrating an envelope correlation coefficient (ECC) below 0.001, high diversity gain (DG), low total active reflection coefficient (TARC), and minimal channel capacity loss (CCL). The combination of a compact footprint, optically reconfigurable band states, wide impedance bandwidth, and outstanding MIMO diversity makes this antenna highly suitable for modern broadband and multi-standard wireless communication systems

Keywords: SWB, MIMO, Pattern Diversity, Microstrip Antenna

1 Introduction

An increasing number of wireless communication devices require compact antenna systems with a higher data transfer rate. MIMO (Multiple Input Multiple Output) antenna systems provide high channel capacity by deploying multiple antennas together for transmission to achieve high gain. Although this technology has already been applied in 4G standards, it will play a vital role in 5G standards, since the installation of a large number of elements will lead to inevitable mutual coupling. MIMO antennas are made reconfigurable by using switches from PIN diodes, varactor diodes or photoconductive switches to operate in multiple frequency bands with different values of gain and isolation. Reconfigurable MIMO antenna systems have the advantage of supporting multipurpose capabilities with high gain and bandwidth. Further, they are compact and compatible with integrated circuits.

Several designs have been proposed to construct reconfigurable MIMO antenna systems for operating in the 5G band over the years. Dual Polarization microstrip MIMO antennas are designed for 5G applications. 8 different fed microstrip antenna elements are arranged on the back of the mobile phone. The measured operating band of -6 dB impedance is 4.37-5.5 GHz and covers the N79 band (4.4-5 GHz) [1]. After then another ultrawideband 4x4 MIMO antenna constructed by two planar inverted F pairs (PIFA pairs) for 5G mobile devices. Two identical PIFA pairs with inverted T shaped open slots are merged as a PIFA-pair with a wide bandwidth of 78% ranging from 3.3 GHz to 7.5 GHz to cover the 5G NR frequency bands: n77/n78/n79 and LTE band 46 [2]. A Wideband Decoupling of Integrated Slot antenna pairs are also used for 5G smartphones. The slot antenna pair with a compact footprint of 28x7x1.8 mm³ shows isolation and performance across 3.3-5.0 GHz. Then, an 8x8 MIMO system, constituted of four sets of slot antenna pairs, was simulated, fabricated, and experimented. This offers better isolation than 10.8 dB, ECCs less than 0.14 between all ports across 3.3-5.0 GHz [3]. Another Compact Multiport MIMO antenna of 4 antenna slots at the corners of antenna ground structure. Each of them has two feeding ports making it a total of eight port antenna. The minimum bandwidth attained by all the ports for S₁₁<10 dB is more than 1.4 GHz from 2.4 GHz to more than 3.8 GHz. The isolation obtained between different ports has a minimum value of -13 dB and the maximum value goes below -30 dB. The ECC is found to be below 0.03 and the peak gain is in the range of 3.2-5 dB [4]. A corrugated Vivaldi phased array antenna in 28 GHz band is proposed in 5G communication. This consist of 4x4 array configuration where the dimension of the elements is 6.46 x 6.46 x 14.25 mm³. The corrugation method is applied to minimize surface current. The overall antenna dimensions are 25.84 x 25.84 x 14.25 mm³. The range of the bandwidth lies between 26-30 GHz, with beam steering capabilities from -30 to +30 degrees. [5]. A MIMO antenna system is designed in a Substrate Integrated Cavity (SIW) to enhance the bandwidth by 80 to 230 MHz [8]. A triband shared-aperture base station antenna array with three wide bands, covering 5G mid-band and the 6G centimetric band. The antenna is not the conventional tube radiator rather than a spiral which shows reduced radar cross section over a range of 4.7-21.5 GHz. The serial resonators help in impedance matching of the segmented-spiral based dipole antenna is achieved across the low band (LB) of 3.05-4.68 GHz, middle band (MB) of 6.2-10 GHz and for high band (HB) of 10-15.6 GHz. The 5G

band spans across 3.3-4.2 GHz. The triband antenna array isolation between any two ports exceeds over 20 dB [6]. A Reconfigurable Structure Reutilization Low-SAR MIMO Antenna for a 4G/5G full-screen metal-frame smartphone with a narrow ground clearance of 2 mm with a LTEAS (long term evolution antenna structures) for the 4G communication system are devised. It is fully functional over 698-960 MHz (LTE low band) and 1710-2690 MHz (LTE high band). The 5G antenna structure cover the spectrum of 3300-4500 MHz [7]. Another example of an antenna of two radiating elements is placed diagonally opposite to each other and connected to a 50-ohm line to achieve a gain of 4.2 dBi and an isolation greater than 12dB in the LTE and early 5G band [9]. With further advancement, MIMO antennas are proposed for 5G sub-6GHz band (3.4-3.8GHz) for future 5G communications [10]. However, most reported works cited here, either focuses on limited bandwidth or lower port count. Therefore, an eight-port compact SWB structure with enhanced isolation is proposed in this paper.

In this paper, a compact reconfigurable eight-port superwideband MIMO antenna is proposed for broadband wireless communication applications. The design employs microstrip line-fed half-circular radiating elements with rotational symmetry and a modified partial ground plane to achieve wide impedance bandwidth, high isolation, and effective pattern diversity over 2–20 GHz. The MIMO performance is evaluated using standard diversity metrics, confirming the suitability of the proposed antenna for high-capacity wireless systems. Ansys High-Frequency Structure Simulator (HFSS) is used to verify the performance of the proposed antenna in software environment.

2 Antenna Design

The proposed antenna's unit cell configuration is shown in Fig. 1. Fig. 1(a) illustrates the top radiating section, while Fig. 1(b) shows the ground plane section.

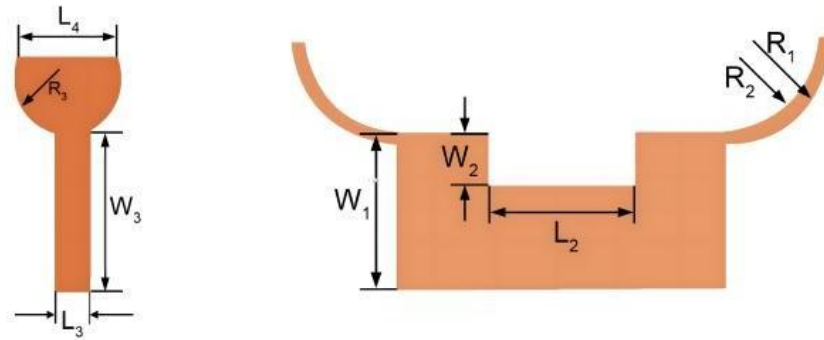


Fig 1. Geometry of the unit cell of proposed reconfigurable MIMO antenna: (a) Top view and (b) Bottom view.

A rectangular aperture is cut from the partial ground plane to get the desired bandwidth. Eight such antenna elements are arranged on a 1.6 mm thick FR-4 substrate

in a rotationally symmetric configuration. The radiating patch and ground plane are put together using a thick 0.035 mm copper layer. Fig. 2 shows the layout of the suggested eight-port SWB-MIMO antenna for wireless applications.

Fig. 3 illustrates the design evolution of the antenna element in three stages: Step I, Step II, and the proposed structure. Step I shows the initial design of a circular microstrip patch fed by a rectangular microstrip line and partial ground plane. The result gives poor impedance matching. Then, Step II introduces a $5 \times 10 \text{ mm}^2$ slot in partial ground plane, leading to better impedance matching and bandwidth. Next, in the proposed structure the circular patch is changed into half circular patch, which improves the impedance matching and results in wide -10dB IBW (Impedance Bandwidth) over 2-20 GHz.

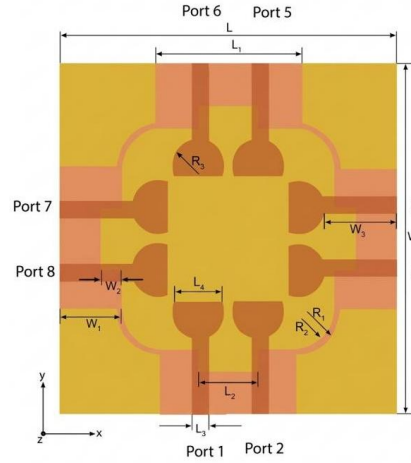


Fig 2. Geometry & design parameters of the proposed eight port SWB- MIMO antenna: [$L1=24\text{mm}$, $L2=10\text{mm}$, $L3=2.7\text{mm}$, $L4=10 \text{ mm}$, $W1= 10 \text{ mm}$, $W2=5 \text{ mm}$, $W3=12.5\text{mm}$, $R1=7.2\text{mm}$, $R2=6.5\text{mm}$, $R3=4.2\text{mm}$]

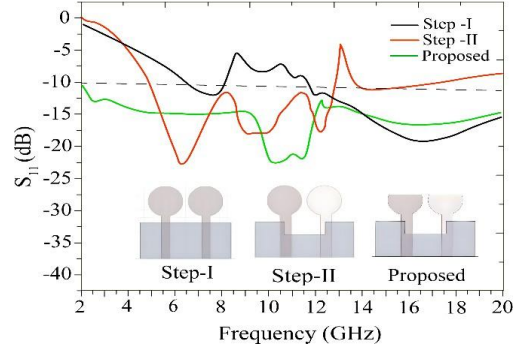


Fig 3. Design advancement of the proposed antenna: (a) Step I, (b) Step II, (c) proposed structure, and corresponding reflection coefficients.

3 Results And Discussion

3.1 S-Parameter Performance

Fig. 4 exhibits S-parameter performance of the suggested eight-port SWB-MIMO antenna. Fig. 4(a) shows reflection coefficients of all eight ports. The -10 dB wider impedance bandwidth extends from 2–20 GHz ($> 180\%$ fractional bandwidth). Fig. 4(b) presents the transmission coefficients for proposed antenna between the ports, which remains below -15 dB over most of the operating band, indicating low mutual coupling between antenna elements.

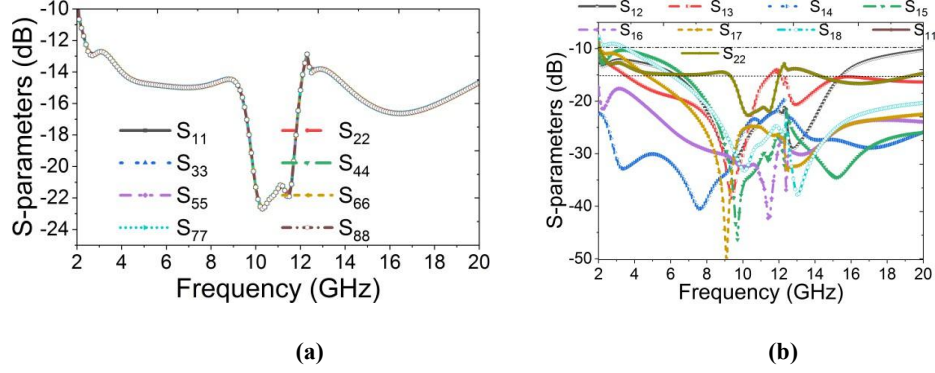


Fig 4. (a) Reflection and (b) Transmission coefficients (simulated) of the proposed eight-port SWB-MIMO antenna.

ECC, DG and CCL parameters are calculated for evaluating the diversity of recommended SWB MIMO Antenna System. The degree of independence between antenna elements evaluated by means of ECC, and derived from the antenna's S-parameters as expressed below:

$$ECC = \frac{|S_i^* S_j + S_{ij}^* S_{ji}|}{(1 - |S_i|^2)(1 - |S_j|^2)} \quad (1)$$

$$DG = 10\sqrt{1 - ECC^2} \quad (2)$$

Equation (1) shows how to calculate the ECC value of an antenna. Where ECC_{ij} is the ECC of the i^{th} , j^{th} antenna elements. ECC variation of the presented SWB-MIMO antenna over the operating frequency range shown in Fig. 5. From this figure, we can see that the ECC value remains below 0.001, indicating very low correlation and excellent diversity.

The DG (Diversity Gain) is another important factor that reflects the improvement in signal reliability. Equation (2) shows how to calculate diversity gain of an antenna. The DG variation of the presented antenna is depicted in Fig. 6. The DG value remains near 10 dB across the operating band, which yields better diversity for the MIMO system.

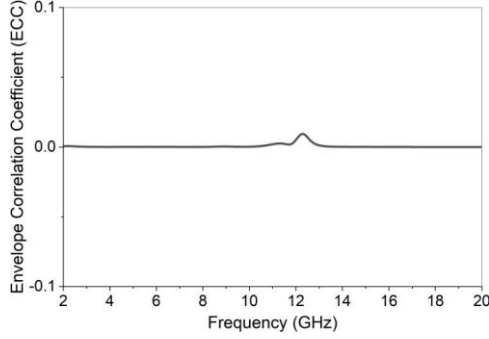


Fig 5. ECC variation of the proposed SWB-MIMO antenna

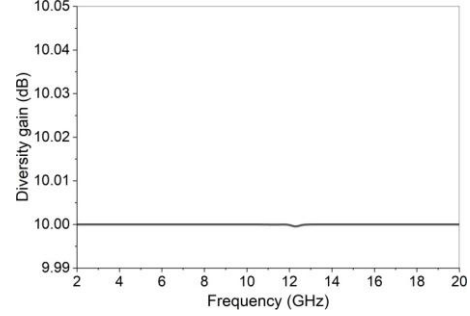


Fig 6. DG variation of the proposed SWB-MIMO antenna

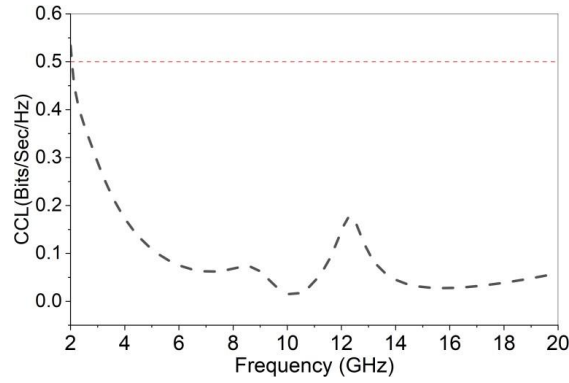


Fig 7. Channel capacity loss (CCL) versus frequency for the proposed reconfigurable SWB-MIMO antenna

The loss in data transmission capacity because of correlation between antenna elements is evaluated by means of Channel Capacity Loss (CCL). Fig. 7 shows the CCL variation of the suggested SWB-MIMO antenna, which is kept under 0.2 across the operating band, indicating reliable performance of MIMO antenna.

3.2 Pattern Diversity Performance of the proposed MIMO Antenna

Pattern diversity is an important characteristic of MIMO antenna systems, as it improves signal reliability in multipath propagation environments by providing multiple independent radiation paths. The radiating elements are placed in an orthogonal orientation, which forms a symmetrical arrangement and due to that we get better pattern diversity, resulting in distinct characteristics of radiation for different antenna ports and reduced spatial correlation. Fig. 8 illustrates the simulated far-field radiation patterns at 10 GHz. Fig. 8(a)–(h) correspond to the radiation patterns obtained when Port 1-8 is excited individually, while the other ports terminate with matched loads. It can be observed that the main radiation lobes of different ports are oriented in different spatial directions with varying field distributions. This directional

variation confirms the existence of effective pattern diversity in the proposed structure. Such diversity behavior is beneficial for improving link quality & mitigating deep fading effects in rich scattering environments.

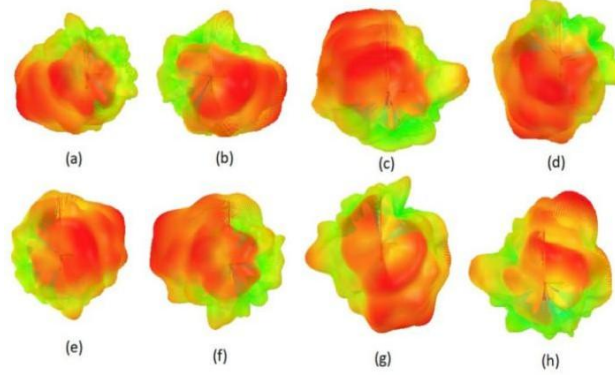


Fig 8. 3D simulation of the radiation patterns shown by our presented SWB-MIMO antenna at 10 GHz in different port excitations.: (a) Port 1, (b) Port 2, (c) Port 3, (d) Port 4, (e) Port 5, (f) Port 6, (g) Port 7, and (h) Port 8, while the other ports terminate with matched loads

3.3 Current Distribution in the proposed antenna

Fig. 9 shows surface current distribution of the suggested SWB-MIMO antenna at 10 GHz when Port 2 is excited, while the other ports terminate with matched loads. By observation, the surface current mainly concentrates around the element which is excited, with minimal current distribution on the other antenna elements, indicating low mutual coupling between the ports. The minimum surface current is 0.055 A/m (represented in blue), while the maximum surface current reaches 88.932 A/m (represented in red).

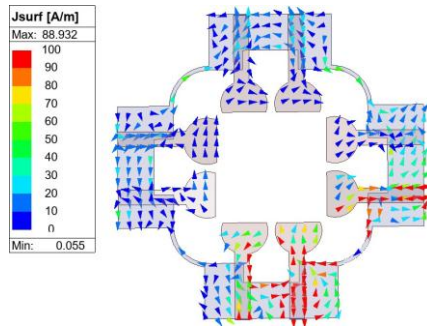


Fig 9. Surface current distribution of the proposed SWB-MIMO antenna at 10 GHz for Port 2 excitation, while other ports are terminated with matched loads

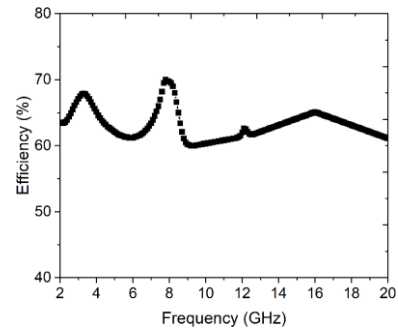


Fig 10. Simulated radiation efficiency versus frequency of the proposed SWB-MIMO antenna

Fig. 10 shows the radiation efficiency of the proposed SWB- MIMO antenna. The efficiency reaches a maximum of approximately 70% across the band of operation. Efficiency has been decreased at certain frequencies by using a lossy FR4 substrate. However, the achieved efficiency remains acceptable for practical applications.

4 Conclusion

A compact eight-port super wideband MIMO antenna is introduced for the applications in high-capacity wireless communication systems. The proposed design employs microstrip line-fed half-circular radiating elements with rotational symmetry combined with a modified partial ground plane, helps to get wide IBW and high inter-element isolation. Simulation results confirm an operating bandwidth from 2 to 20 GHz, covering the sub-6 GHz band and multiple wireless services. The stable radiation characteristics and effective pattern diversity is ensured by the symmetrical and orthogonal arrangement of the antenna elements. Excellent MIMO exhibition is demonstrated through extremely low ECC (below 0.001), high diversity gain, low channel capacity loss, and acceptable radiation efficiency. Owing to its compact size, wide bandwidth, and favorable diversity characteristics, the proposed antenna is ideal for modern super wideband MIMO wireless communication systems.

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